	IT4010 Research Project – 2026 Jan **Topic Assessment Form** **V2.0**
Project ID:	R26-IM-001

1. Topic (12 words max)

Handheld Augmented Reality storybook framework for primary learners' plant growth understanding & stewardship.

2. Research group the project belongs to

MR - Mixed Reality

3. Specialization of the project belongs to

Interactive Media (IM)

4. If a continuation of a previous project:

Project ID	R26-IM-001
Year	2026

5. Brief description of the research problem including references (500 words max) – references not included in word count.

Children aged 8-10 years cannot understand environmental ideas like photosynthesis and air quality from static text-based learning methods [2] [3]. This creates lack of insight into environment-related matters at a tender age. Children of primary schooling age form the primary target group of people affected by this issue, as at this stage of development, engaging methods should be used so that intangible biological processes can be understood by children effectively [4]. At this stage, educators, as well as parents, lack required tools in environmental education if it is not conducted through appropriate Technology-enabled tools [2]. The existing Virtual Reality Virtual Headsets available in the market create ergonomics and safety issues in children [6] [7], while incompetent Augmented Reality applications create Split-Attention effect resulting in Cognitive Overload [5] [10].

The issue has important social repercussions, since children who do not understand the environmental aspect of things have little chances of exercising ecological integrity once they become adults [8]. From the technical perspective, current environmental AR applications in education do not take into account the aspect of affective computing and visualization driven by emotions [1]. There are also economic ramifications, since current MR technology used in schools and other institutions is too costly and dangerous for children [6]. There are ethical concerns, since children are exposed to dangerous hardware, which affects their well-being in these institutions [7]. Lastly, current ecological education has important repercussions, since children have little chances of being associated with nature conservation [8].

The present research will directly contribute to the attainment of Goal No. 4, Quality Education, by fostering equitable, quality, and innovative education through AR technology for children in need of accessible educational tools [3]. Another goal to which this project contributes is Goal No. 13, Climate Action, by fostering awareness and feelings of empathy with nature among children who grow up to be active citizens in taking care of their planet [8]. Gamified storylines are an intrinsic motivator for behavior change [9].

References

[1] L. Botha, "Investigating empathy as a catalyst for pro-environmental behaviour in children," *Leiden University Student Repository,* 2024.

[2] T. I. Ladykova, "Augmented reality in environmental education: A systematic review," *Eurasia Journal of Mathematics, Science and Technology Education,* p. 20, 2024.

[3] D. Safitri, "Transforming environmental education with augmented reality," *Sustainable Development,* 2025.

[4] E. Öztürk, "Can environmental education supported by augmented reality affect students' environmental knowledge, attitudes, and behaviors?," *Educational Process: International Journal,* 2023.

[5] S. Klingenberg, "The impact of augmented reality on cognitive load and performance: A systematic review," *Journal of Computer Assisted Learning,* pp. 285-303, 2021.

[6] K. Silva, "Effects of virtual reality use in children aged 10 to 12 years," *Frontiers in Virtual Reality,* 2025.

[7] P. Kaimara, "Could virtual reality applications pose real risks to children and adolescents? A systematic review of ethical issues and concerns," *Virtual Reality,* p. 26, 2021.

[8] Y. Li, "Empathy with nature promotes pro-environmental attitudes and behaviors," *PsyCh Journal,* 2024.

[9] R. Camacho-Sánchez, "Fostering ecological awareness and pro-environmental intentions through gamification in high school students," *Frontiers in Education,* 2025.

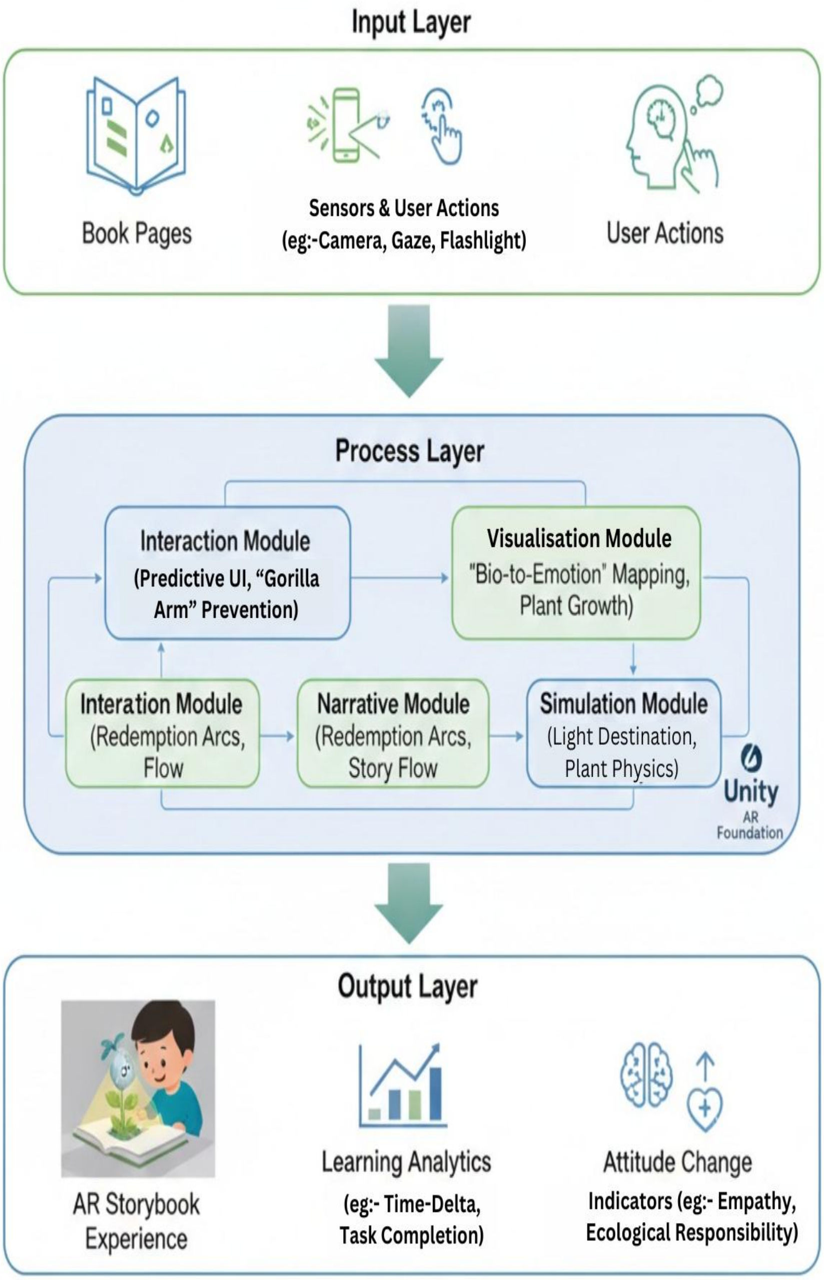
[10] M. Heyder, "The role of cognitive learner prerequisites for cognitive load and learning with augmented reality," *Education Sciences,* p. 14, 2024.

6. Brief description of the nature of the solution including a conceptual diagram (500 words max)

The proposed solution is a handheld Augmented Reality "Magic Book" that runs on standard ARCore/ARKit-enabled smartphones and tablets, allowing school children aged 8-10 in Sri Lanka to explore plant growth and environmental concepts by pointing the device at specially designed storybook pages. Content is based on the local primary school environmental studies syllabus. So that key concepts related to plants parts, growth conditions, pollution, and conservation are consolidated through interactive AR activities rather than static textbook diagrams. From the system perspective, inputs would include printed storybook pages with AR markers or image targets, real-time camera feed, mobile sensor data such as light, gyroscope and accelerometer and user interactions like touch, gestures, and gaze direction inferred from device orientation, supplemented with basic learner profile and progress data.

Within the process layer, four integrated research components operate as coordinated modules. The Hybrid Interaction Framework handles tracking, “phygital” anchoring between physical pages and 3D content, digital signposts to minimize split attention, and occlusion handling for children’s hands. The Affective Visualization module converts plant states into emotion-driven visual and audio cues using procedural growth, aging, and microscopic views, helping 8–10-year-olds emotionally connect with syllabus topics like healthy vs. stressed plants. The Pedagogical Narrative engine manages branching storylines, Tamagotchi-style time-delta calculations, and AI-driven dialogue that responds to plant mood and user behavior, mapping directly to curriculum objectives on responsibility and care for living things. The Context-Aware Simulation engine maps sensor readings such as ambient light and device movement into dynamic growth parameters, including a flashlight-based “grow light” and wind/water simulations that concretize abstract environmental processes.

The output targets emotionally engaged AR experiences that involve virtual plants that grow, wilt, or flourish as a function of the child’s activity, real-world conditions, and their achievement of Grade 3-5 environmental education standards, as well as supportive feedback, rewards, or story-related consequences that foster ecological stewardship.


Input Layer
Book Pages
Sensors & User Actions
(eg:-Camera, Gaze, Flashlight)
User Actions
Process Layer
Interaction Module
(Predictive UI, "Gorilla Arm" Prevention)
Visualisation Module
"Bio-to-Emotion" Mapping, Plant Growth)
Interation Module
(Redemption Arcs, Flow
Narrative Module
(Redemption Arcs, Story Flow
Simulation Module
(Light Destination, Plant Physics)
Unity
AR Foundation
Output Layer
AR Storybook Experience
Learning Analytics
(eg:- Time-Delta, Task Completion)
Attitude Change
Indicators (eg:- Empathy, Ecological Responsibility)

7. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise:

- Expertise in the area of early childhood environmental education through an examination of the current body of knowledge within academic scholarship on learning with AR/XR technology and other topics associated with environmental literacy and learning.
- The guidance will be sought through consultations with teachers of primary schools to ensure the learning outcomes are age-appropriate.
- Specialist knowledge from areas of child psychology and human computer interaction will be used to ensure safe interaction and methods of emotional engagement.

Technical Knowledge & Skills:

- The team members will be able to develop their technical skills through practical development with the use of Unity 3D (2024 LTS), AR Foundation, ARCore, and ARKit.
- Online Courses, Official Documents, and Tutorials related to Mobile AR, Sensors, and Gesture Interaction would be utilized for improvement in technical expertise.
- Prototyping, iteration, and testing of the product would improve the skills acquired in the areas of UI/UX designing, affective visualization, and processing of sensor data
- Peer learning in this team will facilitate sharing of experiences and learnings in development of AR, storytelling, and interaction design aspects.

Data Requirements:

- Data gathering will be performed through observation of children aged 8–10 interacting with the system. While the children are led to believe the technology is acting on its own, their engagement will be evaluated, when in fact, behind the scenes, there is a researcher driving it manually.
- Both pre-test and post-test questionnaires would be used to determine learning outcomes and environmental awareness.
- System data like interaction logs and plant growth status would help with performance analysis.
- Secondary data will also be obtained from existing research literature and curriculum documents.

Ethical Clearance:

- The research requires ethical clearance since it involves children as participants.
- They include parental consent, child assent, anonymization of data, and compliance with ethical requirements for institutions guide the research.

8. Objectives and Novelty

Main Objective Design, develop, and evaluate a child-centered Augmented Reality interactive storybook that enables intuitive, accessible, and engaging learning of plant growth concepts for children aged 8–10 through real-world interaction.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22641410 Wijayarathne K.N.	Develop a hybrid interaction framework that seamlessly bridges physical storytelling with Augmented Reality visuals.	1. Develop a dynamic UI system, which will automatically resize buttons in real time, analyzing the intensity of the device's shake to make up for limitations in the fine motor skills of children. 2. Develop Physical Anchors and an occlusion handling algorithm to provide the AR system with the ability to detect hand gestures and tactile touch on the physical book, without occluding virtual overlays. 3. The flow of interactions is designed to be done in brief, high-impact spurts that are specifically engineered to	A hybrid interaction framework that incorporates predictive UI, dynamic button resizing, physical book-related interaction triggers, as well as fatigue-related interaction design to promote usability and comfort for the child in handheld AR learning.

		prevent the Gorilla Arm effect and physical fatigue in long learning sessions. 4. Conduct usability testing to gauge cognitive load, engagement levels, and effectiveness of the hybrid framework in reducing the split attention effect in the target age group.	
IT22179258 Palliyaguruge S.N.	Design an Affective Visualization Model that translates biological plant states into emotion-responsive visual cues.	1. Design AR optimized 3D plant assets by studying growth stages and children's visual preferences, translating biological conditions into age-appropriate visual cues. 2. Design animations that convey health, stress, and recovery, observing how motion and timing affect children's understanding. 3. Experiment with visual effects such as color shifts, glow effects, audio cues and motion to enhance emotional readability.	Emotion-driven AR plant visualization via procedural aging, microscopic details, empathetic cues, and sound support in emphasizing kids' ecological knowledge.

		4. Conduct child-centered AR trials to assess interpretation, empathy, and refine asset performance.	
IT22237804 Amarawardhane A.S.	Formulate a gamified narrative structure based on behavioral psychology that integrates storytelling with ecological rewards.	1. Create a story for a child with a branch system in which the child's choices will irreversibly affect the growth, mood, and eventual storyline outcome of the plant. The idea should be based on an underlying AI rule-based system to generate dynamically the most appropriate dialogues and feedback, depending on the plant conditions and the environmental context. 2. Create a time-delta-based persistence mechanism that empowers the virtual plant to grow, decay, or recover on its own when the application is inactive in order to better simulate real biological processes. 3. Introducing non-point rewards through narrative consequences, and developing a conservation action that involves redemption and allows the child an opportunity for rectification of	A gamify story engine that integrates artificial intelligence-powered conversations, dynamic engagement mechanisms, ever-present plant vitality systems, and conservation efforts given for redemption.

		their mistakes in hopes of improving plant health. 4. Evaluate changes in environmental attitudes and feelings of responsibility in children through pre- and post-interaction evaluation tools.	
IT22626974 Dissanayaka D.M.D.D.	Design a context-aware simulation engine that maps physical environmental data information in virtual parameters.	1. Develop a system that will help the app determine the amount of light in the environment, so that the app could know the brightness and dimness of the environment. 2. Create an algorithm to determine how to simulate plant growth based upon light exposure, climate conditions, and various interactions managed by users. 3. Enhance the aggregation of data from various sensors so the app runs without a hitch without excessively consuming battery life. 4. Develop a feature that can turn on the device's flashlight whenever it senses that it is too dark in the	Innovative interactions by using the device flashlight as a virtual grow light with inverse intensity, mapping real-world environmental changes to the virtual plant, and leveraging the gyroscope for wind and water simulations.

		surroundings and assist in allowing the virtual plant to grow.	

9. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Ishara	Gamage	
Co-Supervisor	Mr.	Nushkan	Nizmi	
External Supervisor				
Summary of external supervisor’s (if any) experience and expertise				